

Lecture 7

Wave equation

in $\mathbb{R}^n$. $\Delta = \sum_{j=1}^n \partial_{x_j}^2$

$(1+n)$ -dimensional (homogeneous) wave equation:

$$(\partial_t^2 - \Delta)u = 0, \quad u: \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}$$

$$\text{i.e. } \partial_t^2 u(t, x) - \sum_{j=1}^n \partial_{x_j}^2 u(t, x) = 0$$

Rmk. This is a linear equation

We wish to understand the Cauchy problem:

$$\left\{ \begin{array}{l} (\partial_t^2 - \Delta)u = 0 \\ u(0, x) = g(x) \\ \partial_t u(0, x) = h(x) \end{array} \right\} \text{ initial data}$$

Rmk. wave eq is second order in time, so we need two pieces of initial data, analogous to how second order ODEs need both initial position and momentum to be well-posed.

1+1 Wave equation

$$(IVP) \begin{cases} (\partial_t^2 - \partial_x^2) u = 0 \\ u(0, x) = g(x) \\ \partial_t u(0, x) = h(x) \end{cases}$$

Thm. for $g \in C^2(\mathbb{R})$ and $h \in C^1(\mathbb{R})$ (IVP) admits a unique solution

$$u(t, x) = \frac{1}{2} [g(x+t) + g(x-t)] + \frac{1}{2} \int_{x-t}^{x+t} h(\tau) d\tau$$

Idea. factor $\partial_t^2 - \partial_x^2 = (\partial_t + \partial_x)(\partial_t - \partial_x)$
first order operators

Pf. define $w := \partial_t u - \partial_x u$, then

$$\begin{cases} \partial_t w + \partial_x w = 0 \\ w(0, x) = w_0(x) \end{cases}$$

← we will come back and figure out what this is later

Method of characteristics $\Rightarrow w(t, x) = w_0(x-t)$

Rmk. Recall the characteristic lines for this was given by $x(t) = x_0 + t$, $x_0 \in \mathbb{R}$.

Now we solve

$$\begin{cases} \partial_t u - \partial_x u = w \\ u(0, x) = g(x) \end{cases}$$

Characteristics are $x_-(t) = x_0 - t$

$\leadsto$ along characteristics, u satisfies

$$\begin{cases} \frac{d}{dt} u(t, x_0 - t) = w(t, x_0 - t) \\ u(0, x_0) = g(x_0) \end{cases}$$

$$\Rightarrow u(t, x_0 - t) = g(x_0) + \int_0^t w(s, x_0 - s) ds$$

$$\begin{aligned} \text{so } u(t, x) &= g(x+t) + \int_0^t w(s, x+t-s) ds \\ &= g(x+t) + \int_0^t w_0(x+t-2s) ds \end{aligned}$$

What is w_0 ? recall

$$\begin{aligned} w_0(x) &= \partial_t u(0, x) - \partial_x u(0, x) \\ &= h(x) - \partial_x g(x) \end{aligned}$$

Then

$$u(t, x) = g(x+t) + \int_0^t h(x+t-2s) - g'(x+t-2s) ds$$

$$\boxed{\tau = x+t-2s} = g(x+t) + \frac{1}{2} \int_{x-t}^{x+t} h(\tau) - g'(\tau) d\tau$$

$$= \frac{1}{2} [g(x+t) + g(x-t)] + \frac{1}{2} \int_{x-t}^{x+t} h(\tau) d\tau \quad \square$$

Ex. $h(x) = 0$

$t=1$



$t=2$



$t=3$



Another perspective

Change coordinates to characteristic coordinates.

Define $\xi = t + x$ $\eta = t - x$

i.e. $t = \frac{\xi + \eta}{2}$ $x = \frac{\xi - \eta}{2}$

$$\Rightarrow \partial_{\xi} = \frac{1}{2}(\partial_t + \partial_x) \quad \partial_{\eta} = \frac{1}{2}(\partial_t - \partial_x)$$

Note then $(\partial_t^2 - \partial_x^2)u = 0 \iff 4\partial_{\xi}\partial_{\eta}\tilde{u} = 0$

where $\tilde{u}(\xi, \eta) = u(t(\xi, \eta), x(\xi, \eta))$

Rmk. Ignoring initial condis and integrating

$\partial_{\xi}\partial_{\eta}\tilde{u} = 0$, we see that

$$\tilde{u}(\xi, \eta) = V_+(\xi) + V_-(\eta)$$

for some $V_{\pm} : \mathbb{R} \rightarrow \mathbb{R}$. Changing coords, we see

$$u(t, x) = V_+(x+t) + V_-(x-t)$$

Check that the solution formula takes this form

Let's try to solve 1+1 wave in these coords.
 this time we add an inhomogeneity (i.e. a forcing term).

$$(\text{IVP}) \begin{cases} (\partial_t^2 - \partial_x^2) u = f(t, x) \\ u(0, x) = 0 \\ \partial_t u(0, x) = 0 \end{cases} \quad f \in C^0([0, \infty) \times \mathbb{R})$$

what are the initial cond's in $(\tilde{z}, \eta)$ coordinates?

$$u(0, x) = \tilde{u}(x, -x) = 0$$

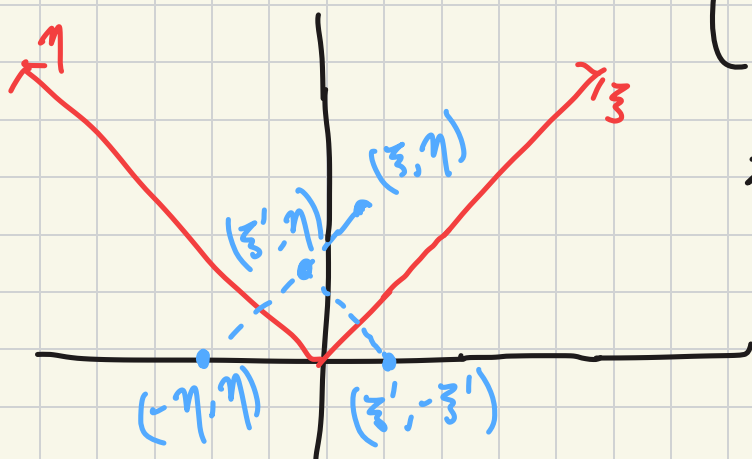
$$\partial_t u(0, x) = (\partial_{\tilde{z}} + \partial_{\eta}) \tilde{u}(x, -x) = 0$$

$$\partial_x u(0, x) = (\partial_{\tilde{z}} - \partial_{\eta}) \tilde{u}(x, -x) = 0$$

$$\Rightarrow \partial_{\tilde{z}} \tilde{u}(x, -x) = 0$$

(IVP) equivalent to

$$\begin{cases} \partial_{\eta} \partial_{\tilde{z}} \tilde{u} = \frac{1}{4} \tilde{f} \\ u(\eta, -\eta) = 0 \\ \partial_{\tilde{z}} \tilde{u}(\eta, -\eta) = 0 \end{cases}$$



$$\begin{aligned} \tilde{u}(\tilde{z}, \eta) &= \int_{-\eta}^{\tilde{z}} \partial_{\tilde{z}} \tilde{u}(\tilde{z}', \eta) d\tilde{z}' \\ &\quad + \tilde{u}(-\eta, \eta) \\ &= \int_{-\eta}^{\tilde{z}} \partial_{\tilde{z}} \tilde{u}(\tilde{z}', \eta) d\tilde{z}' \end{aligned}$$

$$\partial_{\xi} \tilde{u}(\xi, \eta) = \int_{-\xi}^{\eta} \partial_{\eta} \partial_{\xi} \tilde{u}(\xi', \eta') d\xi' d\eta' + \partial_{\xi} \tilde{u}(\xi, -\xi)$$

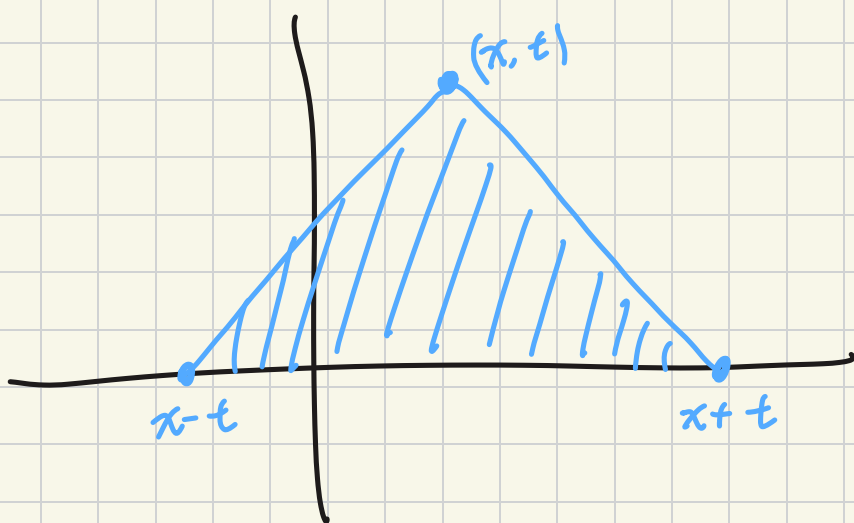
$$= \frac{1}{4} \int_{-\xi}^{\eta} \tilde{f}(\xi', \eta') d\xi' d\eta'$$

$$\Rightarrow \tilde{u}(\xi, \eta) = \frac{1}{4} \int_{-\eta}^{\xi} \int_{-\xi'}^{\eta} \tilde{f}(\xi', \eta') d\xi' d\eta'$$

C.O.V. back to (t, x)

$$d\xi' d\eta' = \left| \frac{\partial(\xi', \eta')}{\partial(t', x')} \right| dt' dx'$$

$$= \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} dt' dx' = 2 dt' dx'$$



$$\tilde{u}(t, x) = \frac{1}{2} \int_0^t \int_{x-(t-t')}^{x+(t-t')} f(t', x') dt' dx'$$

Thm (d'Alembert's formula).

$$\begin{cases} (\partial_t^2 - \partial_x^2) u = f \\ u(0, x) = g(x) \\ \partial_t u(0, x) = h(x) \end{cases}$$

$$\begin{aligned} u(t, x) = & \frac{1}{2} \int_0^t \int_{x-(t-t')}^{x+(t-t')} f(x', t') dx' dt' \\ & + \frac{1}{2} [g(x+t) + g(x-t)] \\ & + \frac{1}{2} \int_{x-t}^{x+t} h(\tau) d\tau \end{aligned}$$